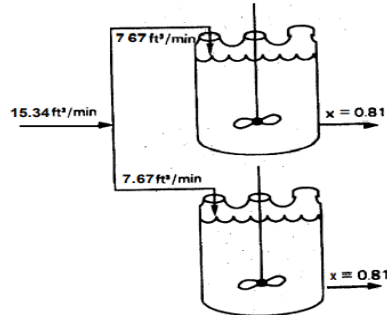


|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                       |           |       |        |       |      |     |          |        |      |      |      |       |       |       |      |     |          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------|-------|--------|-------|------|-----|----------|--------|------|------|------|-------|-------|-------|------|-----|----------|
|                       | <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Winter Examination – 2022</b><br><b>Course: B. Tech.                      Branch: Chemical                      Semester: V</b><br><b>Subject Code &amp; Name: BTCHC502 Chemical Reaction Engineering- I</b><br><b>Max Marks: 60                      Date: 31/01/2023                      Duration: 3 Hr.</b>                                                                                                                                                                                                                                                                                                                                                                                               |                       |           |       |        |       |      |     |          |        |      |      |      |       |       |       |      |     |          |
|                       | <b>Instructions to the Students:</b><br>1. All the questions are compulsory.<br>2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.<br>3. Use of non-programmable scientific calculators is allowed.<br>4. Assume suitable data wherever necessary and mention it clearly.                                                                                                                                                                                                                                                                                                                                                                                                 |                       |           |       |        |       |      |     |          |        |      |      |      |       |       |       |      |     |          |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | (Level/CO)            | Marks     |       |        |       |      |     |          |        |      |      |      |       |       |       |      |     |          |
| <b>Q. 1</b>           | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       | <b>12</b> |       |        |       |      |     |          |        |      |      |      |       |       |       |      |     |          |
| <b>A)</b>             | Derive and explain with assumptions and characteristics general mole balance equation of a Plug Flow Reactor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CO1                   | <b>6</b>  |       |        |       |      |     |          |        |      |      |      |       |       |       |      |     |          |
| <b>B)</b>             | A liquid phase reaction $A + B \longrightarrow 2R$ is carried out in a flow reactor. The feed concentrations are $C_{A0}$ =100 mol/it and $C_{B0}$ = 150 mol/lit. If conversion of reactant A is 90 % then find exit concentrations of reactants A, B and product R. (Find $C_A$ , $C_B$ , $C_R$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | CO1                   | <b>6</b>  |       |        |       |      |     |          |        |      |      |      |       |       |       |      |     |          |
| <b>C)</b>             | For a isothermal gas phase isomerization reaction $A \longrightarrow B$ , the laboratory measurements are given in the table below which shows the chemical rate as a function of conversion. The temperature was 500K and the total pressure was 830kPa. The initial charge to the reactor was pure A.<br><table border="1"><tr><td>X</td><td>0</td><td>0.1</td><td>0.2</td><td>0.4</td><td>0.6</td><td>0.7</td><td>0.8</td></tr><tr><td><math>-r_A</math></td><td>0.45</td><td>0.37</td><td>0.30</td><td>0.195</td><td>0.113</td><td>0.079</td><td>0.05</td></tr></table><br>If the reaction is carried out in a PFR, what is the PFR volume necessary to achieve 80% conversion if entering molar flow rate $F_{A0}$ = 0.4 mol/s ? Use Simpson's five-point quadrature formula. | X                     | 0         | 0.1   | 0.2    | 0.4   | 0.6  | 0.7 | 0.8      | $-r_A$ | 0.45 | 0.37 | 0.30 | 0.195 | 0.113 | 0.079 | 0.05 | CO1 | <b>6</b> |
| X                     | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.1                   | 0.2       | 0.4   | 0.6    | 0.7   | 0.8  |     |          |        |      |      |      |       |       |       |      |     |          |
| $-r_A$                | 0.45                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.37                  | 0.30      | 0.195 | 0.113  | 0.079 | 0.05 |     |          |        |      |      |      |       |       |       |      |     |          |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                       |           |       |        |       |      |     |          |        |      |      |      |       |       |       |      |     |          |
| <b>Q.2</b>            | <b>Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       | <b>12</b> |       |        |       |      |     |          |        |      |      |      |       |       |       |      |     |          |
| <b>A)</b>             | Show for what conditions and manipulation the expression for $C_B$ for a gas flow system reduces to $C_B=C_{A0}[\frac{\theta B-(\frac{b}{a})X}{(1+\epsilon X)}] \left(\frac{P}{P_0}\right)\left(\frac{T_0}{T}\right)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CO2                   | <b>6</b>  |       |        |       |      |     |          |        |      |      |      |       |       |       |      |     |          |
| <b>B)</b>             | Draw a stoichiometric table for a flow system having a reaction;<br>$aA+ bB \longrightarrow cC+ dD$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CO2                   | <b>6</b>  |       |        |       |      |     |          |        |      |      |      |       |       |       |      |     |          |
| <b>C)</b>             | The rule of thumb that the rate of reaction doubles for a 10°C increase in temperature occurs only at a specific temperature for a given activation energy. Determine the activation energy and frequency factor from the following data:<br><table border="1"><tr><td>K (min)<sup>-1</sup></td><td>0.001</td><td>0.05</td></tr><tr><td>T (°C)</td><td>0</td><td>100</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                             | K (min) <sup>-1</sup> | 0.001     | 0.05  | T (°C) | 0     | 100  | CO2 | <b>6</b> |        |      |      |      |       |       |       |      |     |          |
| K (min) <sup>-1</sup> | 0.001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.05                  |           |       |        |       |      |     |          |        |      |      |      |       |       |       |      |     |          |
| T (°C)                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 100                   |           |       |        |       |      |     |          |        |      |      |      |       |       |       |      |     |          |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                       |           |       |        |       |      |     |          |        |      |      |      |       |       |       |      |     |          |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                               |                                      |                               |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------|-------------------------------|-----|--------|-----------------------|-----|--------|-----|-----|-------|-----|---|------|------|---|-------|-----|-----|---|
| Q. 3                  | Solve Any Two of the following.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                               | 12                                   |                               |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |
| A)                    | Explain in detail the scale up of laboratory reactor data to design a continuous stirred tank reactor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CO3                           | 6                                    |                               |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |
| B)                    | Explain in detail Space time and Damkolher number for first order and second order system. For two 800-gal CSTRs arranged in parallel with 7.67 ft <sup>3</sup> /min ( $v_o/2$ ) fed to each reactor. Calculate the space time and Damkolher number for the given system where $k= 0.311 \text{ min}^{-1}$ and reaction kinetics is $-r_A = KC_A$ .<br>                                                                                                                                                                                       | CO3                           | 6                                    |                               |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |
| C)                    | The elementary gas phase reaction $A \longrightarrow C + 2D$ is carried out isothermally in a flow reactor with no pressure drop. The specific reaction rate constant at 50 °C is $10^{-4} \text{ min}^{-1}$ and the activation energy is 85 kJ/mol. Pure reactant A enters the reactor at 10 atm and 127 °C and a molar flow rate of 2.5 mol/min. Calculate the reactor volume and space time to achieve 90% conversion in a) PFR and b) CSTR.                                                                                                                                                                                | CO3                           | 6                                    |                               |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                               |                                      |                               |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |
| Q.4                   | Solve Any Two of the following.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                               | 12                                   |                               |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |
| A)                    | The decomposition of $\text{NH}_3$ on a tungsten wire at 856 °C yielded the following data.<br><table border="1" data-bbox="175 1164 1128 1314"><tr><td>Time (sec)</td><td>200</td><td>400</td><td>600</td><td>1000</td></tr><tr><td>Total pressure (torr)</td><td>228</td><td>250</td><td>273</td><td>318</td></tr></table><br>Prove that the reaction is zero order and determine the value of rate constant.<br>Reaction: $2\text{NH}_3 \longrightarrow \text{N}_2 + 3\text{H}_2$                                                                                                                                           | Time (sec)                    | 200                                  | 400                           | 600 | 1000   | Total pressure (torr) | 228 | 250    | 273 | 318 | CO4   | 6   |   |      |      |   |       |     |     |   |
| Time (sec)            | 200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 400                           | 600                                  | 1000                          |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |
| Total pressure (torr) | 228                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 250                           | 273                                  | 318                           |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |
| B)                    | The gas-phase decomposition $A \longrightarrow B+2C$ is carried out in a constant-volume batch reactor. From the data in table below, determine the reaction order and specific reaction rate.<br><table border="1" data-bbox="279 1561 1054 1843"><tr><td>Run</td><td>Initial Conc., <math>C_{A0}</math><br/>(g mol/L)</td><td>Half-life, <math>t_{1/2}</math><br/>(min)</td></tr><tr><td>1</td><td>0.0250</td><td>4.1</td></tr><tr><td>2</td><td>0.0133</td><td>7.7</td></tr><tr><td>3</td><td>0.010</td><td>9.8</td></tr><tr><td>4</td><td>0.05</td><td>1.96</td></tr><tr><td>5</td><td>0.075</td><td>1.3</td></tr></table> | Run                           | Initial Conc., $C_{A0}$<br>(g mol/L) | Half-life, $t_{1/2}$<br>(min) | 1   | 0.0250 | 4.1                   | 2   | 0.0133 | 7.7 | 3   | 0.010 | 9.8 | 4 | 0.05 | 1.96 | 5 | 0.075 | 1.3 | CO4 | 6 |
| Run                   | Initial Conc., $C_{A0}$<br>(g mol/L)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Half-life, $t_{1/2}$<br>(min) |                                      |                               |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |
| 1                     | 0.0250                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4.1                           |                                      |                               |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |
| 2                     | 0.0133                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 7.7                           |                                      |                               |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |
| 3                     | 0.010                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 9.8                           |                                      |                               |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |
| 4                     | 0.05                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1.96                          |                                      |                               |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |
| 5                     | 0.075                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1.3                           |                                      |                               |     |        |                       |     |        |     |     |       |     |   |      |      |   |       |     |     |   |

|      |                                                                                                                                                  |     |      |      |      |     |      |      |     |    |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----|------|------|------|-----|------|------|-----|----|
| C)   | The irreversible isomerization $A \longrightarrow B$ was carried out in a batch reactor and the following concentration-time data were obtained: |     |      |      |      |     |      |      | CO4 | 6  |
|      | t (min)                                                                                                                                          | 0   | 3    | 5    | 8    | 10  | 12   | 15   |     |    |
|      | $C_A$ (mol/dm <sup>3</sup> )                                                                                                                     | 4.0 | 2.89 | 2.25 | 1.45 | 1.0 | 0.65 | 0.25 |     |    |
|      | Determine the reaction order, $\alpha$ and the specific reaction rate, $k$                                                                       |     |      |      |      |     |      |      |     |    |
|      |                                                                                                                                                  |     |      |      |      |     |      |      |     |    |
| Q. 5 | Solve Any Two of the following.                                                                                                                  |     |      |      |      |     |      |      |     | 12 |
| A)   | What is catalyst? Explain different types of surface reactions mechanisms in a catalytic reactions.                                              |     |      |      |      |     |      |      | CO5 | 6  |
| B)   | Draw a neat diagram showing all the steps in a heterogeneous catalytic reaction.                                                                 |     |      |      |      |     |      |      | CO5 | 6  |
| C)   | Explain types of catalyst deactivation.                                                                                                          |     |      |      |      |     |      |      | CO5 | 6  |
|      | *** End ***                                                                                                                                      |     |      |      |      |     |      |      |     |    |

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